SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

MACHINE LEARNING

GENERATIVE AI

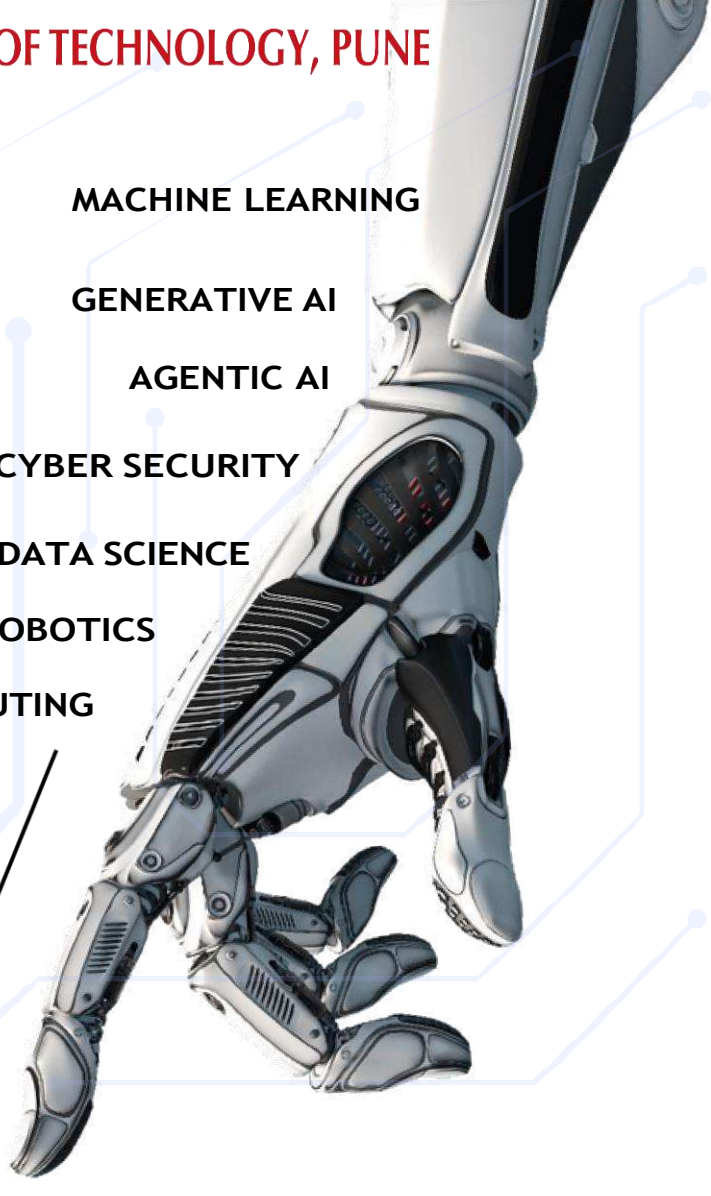
AGENTIC AI

CYBER SECURITY

DATA SCIENCE

ROBOTICS

CLOUD COMPUTING



SYMBI SAINS

ANNUAL DEPARTMENTAL MAGAZINE OF

COMPUTER SCIENCE AND ENGINEERING

EDITION 2025-26

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THE EDITORIAL

Welcome to Symbi-Sains, the annual magazine of the Computer Science and Engineering department at Symbiosis Institute of Technology.

This edition shows the collective efforts, ideas, and progress of our CSE branch. It brings together not just technical knowledge, but also the work students are actively doing, projects they are building, opportunities they are pursuing, and the skills they are developing along the way.
Symbi-Sains continues to serve as a platform where students and faculty can share their experiences and perspectives. This year, we have focused on presenting a more practical and student-driven view of Computer Science, including project showcases, student achievements, career insights, and discussions on emerging technologies.

The aim is simple, to create something that is not only informative, but also relevant. Along with technical articles, this edition highlights real journeys, lessons learned, and the different paths students are taking as they grow in this field.
We believe that learning becomes more meaningful when it is shared. By bringing together these contributions, we hope to encourage collaboration, curiosity, and a mindset of continuous improvement within the department.
As you go through these pages, we hope you find them engaging and useful. More importantly, we hope this edition motivates you to explore, build, and make the most of the opportunities around you.

Have a great read!

Editor-in-Chief
CSE Department Magazine

THE EDITORIAL TEAM

CHIEF EDITOR



DR. SONALI KOTHARI

FACULTY EDITOR



DR. MADHURI HIWALE

EDITOR IN CHIEF



DHARMIT SHAH

EDITOR IN CHIEF



DEVANSH SINGH

MANAGING EDITOR



AKUL AGARWAL

CREATIVE DIRECTOR



NITYA GOEL

VISION AND MISSION

THE VISION

To be a centre of excellence in Computer Science and Engineering that fosters innovation, multidisciplinary learning, and global competence while contributing to sustainable societal development.

THE MISSION

- CSE-M1: To equip students with strong technical foundations, problem-solving abilities, and leadership skills in diverse domains of Computer Science and Engineering.
- CSE-M2: To drive innovation and research through collaboration with academia, industry, and public-private organizations.
- CSE-M3: To nurture ethical, socially responsible, and environmentally conscious computing professionals who create inclusive and sustainable solutions.
- CSE-M4: To cultivate a global outlook by encouraging international exposure, collaborations, and continuous learning in emerging technologies.

PROGRAM EDUCATIONAL OBJECTIVES

- PEO1: Apply their skills with research orientation and establish themselves globally.
- PEO2: Apply problem solving and emerging technology skills for designing solutions.
- PEO3: Apply technical and leadership skills to be a successful entrepreneur

FROM THE DIRECTOR'S DESK



Dr. Ketan Kotecha

Dean, Faculty of Engineering, Professor & Director, Symbiosis Institute of Technology, Head, Symbiosis Centre for Applied Artificial Intelligence (SCAAI)

Welcome to the Department of Computer Science Engineering, where you'll join a diverse community engaged in hands-on learning and cutting-edge research on your route to a successful career in business, government, or academia. Through creative instruction from our professors and involvement in hands-on projects and laboratories, our undergraduate and postgraduate students get an excellent educational experience.

Being a member of one of our student competition teams allows You to acquire not just engineering and interpersonal abilities but also lifelong friendships. Working on industry-sponsored capstone projects and internships in businesses fosters professional ties that lead to post-graduation success.

Our faculty members are engaged in high-impact research funded by government and academic bodies in a variety of areas, including Artificial Intelligence and Machine Learning, Deep Learning, Cyber Security, Blockchain, and Data science. Through excellence in teaching, research, and service, we create an inclusive and dynamic experience. Welcome, I look forward to working with you. If you have any comments, questions, or ideas, please email me at director@sitpune.edu.in.

FROM THE HEAD'S DESK



Dr. Sonali Kothari

Professor and Head, Department of Computer Science & Engineering, SIT Pune

Welcome to the Department of Computer Science and Engineering (CSE) at Symbiosis Institute of Technology, Pune. Computer science Engineering is a fascinating, challenging, and dynamic discipline that is now generally acknowledged as a vital source of tools and strategies for improvements in virtually every field of human effort. The Department of CSE started its journey in 2008 by offering undergraduate and postgraduate programs. Currently, the Department has around 30 skilled full-time faculty members, many of whom are PhD holders.

The department of CSE at SIT is intensively research-oriented and student-centered. Our mission and vision statements express some of our department's guiding concepts and objectives. You will be exposed to high-quality, inspiring teaching and learning as a student in the Department. You'll get the chance to work on real-world issues presented by the industry and learn from IT & international experts/researchers. Students can enhance their experiences and grow personally by participating in and organizing extracurricular activities such as technical events, hackathons, poster presentations, conferences, student clubs, and other activities that promote the development of a variety of both discipline-specific and lifelong transferable skills such as enterprise, collaboration, and leadership.

We are quite proud of our graduates and their achievements; they are highly competent IT professionals who are pursuing successful professions in a variety of settings in a global market. If you have any comments, questions, or ideas, please email me at headcse@sitpune.edu.in.

FROM FACULTY EDITOR'S DESK



Dr. Madhuri Hiwale

Faculty Editor, Department of Computer Science & Engineering, SIT Pune

It gives me immense pleasure to present **Symbi-Sains**, the annual magazine of the Department of Computer Science and Engineering, Symbiosis Institute of Technology, Pune.

This edition captures the spirit of our CSE community through the ideas, achievements, innovations, projects, research, and experiences of our students and faculty. It reflects our commitment to continuous learning, emerging technologies, creativity, and meaningful application of knowledge. In a rapidly evolving technological world, success is not defined only by what we know, but by our ability to think critically, learn continuously, innovate responsibly, and create solutions that make a difference**.**

I sincerely appreciate the efforts of every student, faculty member, and editorial team member who contributed to this edition. May these pages inspire curiosity, celebrate accomplishments, and encourage our readers to explore new possibilities. I would like to congratulate all the students, faculty members, contributors, and members of the editorial team whose efforts have brought this magazine to life. A magazine is more than a collection of articles, it is a record of our journey, a celebration of our achievements, and an inspiration for what lies ahead**.**

I hope that every reader finds something meaningful in these pages an idea that sparks curiosity, an achievement that inspires them to take their next step.

DEPARTMENT OVERVIEW

The Department of Computer Science and Engineering is at the forefront of technological advancement and is dedicated to delivering high-quality, industry-aligned education. Students gain hands-on experience in real-world problem-solving and application development through a range of initiatives such as technical workshops, industry internships, international collaborations, value-added courses, and capstone projects. The department boasts a highly qualified faculty, certified and trained in emerging and advanced technologies including Cloud Computing, DevOps, Full Stack Development, Cyber Security, Artificial Intelligence and Machine Learning, Data Science, Blockchain Technology, Internet of Things (IoT), Big Data Analytics, Augmented and Virtual Reality (AR/VR), and modern frameworks like Spring Boot. Accreditation by the National Board of Accreditation (NBA) reaffirms the department's commitment to upholding rigorous academic standards and fostering excellence in technical education and professional training.

The department is dedicated to the holistic development of students, preparing them for successful careers in industry, government, higher education, and entrepreneurship. Students are encouraged and supported in launching their own start-ups and are provided with platforms to innovate and lead. They gain practical exposure by working on live projects in collaboration with academic mentors and industry experts, which equips them with the skills needed for a smooth transition into professional life. A strong student–faculty interaction culture ensures individual attention and continuous academic support. The curriculum includes a six-month internship, which significantly enhances their industry readiness. Students are actively involved in both application-oriented and research-driven projects, often in collaboration with R&D labs, research centres, industries, and international universities. The department's faculty members are highly competent, actively engaged in research and publication, and have undertaken prestigious projects in partnership with reputed industries and global research institutions.

The department has state of art computing facility such as

- High performance computing lab with Nvidia GPU
- Data Analytics and Visualization lab
- Deep Learning Lab with higher processing power
- Computer Centre with machines in high configuration and licensed software' s

ARTICLES

Programming is a Skill. Computer Science is a Way of Thinking.



DHARMIT KUNAL SHAH

Student, Computer Science & Engineering
B.Tech , 2025-2029

When I entered my first semester of Computer Science and Engineering, my goal was simple: learn how to code well. I wanted my programs to compile without errors. I wanted my logic to work. I wanted an output that matched expectations. And in the beginning, that felt like progress.

But somewhere between writing small programs and analyzing why they worked, I began to notice something important. Programming and computer science are not the same thing. Programming teaches you how to instruct a machine. Computer science teaches you how to think about problems.

The Early Satisfaction of Code

In the first semester, solving a problem usually means getting the correct output. If a sorting function arranges numbers properly, or if a program handles input correctly, the task feels complete. That stage is important. Writing code builds confidence. It improves logical sequencing and forces precision. But correctness is only the surface layer. While implementing a simple solution once, I noticed that two different approaches produced the same output yet one took significantly more time when the input grew larger. That observation changed how I saw coding. The question shifted from "Does it work?" to "How well does it work?"

Where Computer Science Begins

That shift from working code to efficient design is where computer science begins.
Computer science asks deeper questions:
What is the time complexity of this approach?
How does it behave as the data scales?
Is there a more optimal structure?
What are the theoretical limits of this method?
These questions move beyond syntax. They require analysis. When we study data structures, we are not just memorizing definitions. We are learning how information can be organized to reduce inefficiency. When we examine algorithmic complexity, we are not simply performing calculations. We are measuring scalability. This kind of thinking transforms coding into structured reasoning.

A Change in Perspective

Initially, I measured my growth by how much code I could write. Now I measure it differently.
Do I understand why a solution works?
Can I redesign it more elegantly?
Can I explain its limitations clearly?
Those questions feel more aligned with computer science than simply producing output.
And interestingly, asking those questions makes programming itself more meaningful.

What I Learned from This Shift

Programming builds the ability to create. Computer science builds the ability to design. One focuses on execution. The other focuses on structure. Recognizing this difference early even in the first semester changes how you approach learning. It encourages depth over speed. In a field that evolves as quickly as technology does, depth is what allows long-term growth. Languages will change. Frameworks will change. But the ability to think rigorously about problems will always remain valuable.

ARTICLES

Generative AI: A Student's Companion in the Digital Age



DEVANSH SINGH

Student, Computer Science & Engineering
B.Tech , 2025-2029

When people hear the term Generative AI, they often think of chatbots that answer questions or tools that create images and content within seconds. While these features are impressive, my experience with Generative AI has been much more meaningful. For me, it has become a learning tool that supports me through academics, technology, and everyday problem-solving.

As a Computer Engineering student, When I encounter topics that can seem difficult at first. Whether it is programming, software engineering concepts, or exam preparation, there is always something new to learn. Earlier, finding the right explanation often meant searching through multiple websites, videos, and notes. With Generative AI, I can ask questions in a simple way and receive explanations that are easy to understand. It feels less like using a search engine.

One of the biggest advantages of AI has been during exam preparation. Like many students, I have faced situations where a large syllabus had to be covered in a short time. AI helped me organize topics into concise notes and summaries, making revision more efficient. Instead of spending hours collecting information through pdf/ppts, I could focus on understanding concepts and strengthening my fundamentals.

As someone interested in programming, I also appreciate how AI simplifies debugging and problem-solving. Every programmer has faced situations where an error message appears confusing or a solution seems impossible to find. AI can explain technical issues in simple language and suggest possible approaches to solve them. Even when the first suggestion does not work, the discussion often provides new insights and helps me think more logically about the problem. This not only saves time but also strengthens analytical and problem-solving skills.

GENERATIVE AI & STUDENTS

- Learn Faster
- Understand Concepts Better
- Solve Programming Problems
- Explore New Technologies
- Troubleshoot Technical Issues
- Continuously Improve Skills

Another area where AI has helped me is exploring new technologies. During my transition from Windows to Fedora GNOME (Workstation), I encountered several challenges while learning a completely different environment. AI served as a guide that helped me understand Linux tools, troubleshoot issues, and become more confident in experimenting with a new operating system environment. Rather than simply providing answers, it encouraged me to learn how things work behind the scenes. Of course, Generative AI is not perfect. There have been occasions when suggested solutions did not work or when the information provided required manual verification. These experiences reminded me that AI should be treated as a helpful assistant rather than an accurate source. Critical thinking, and independent learning remain just as important as ever. The real value of AI lies in supporting human intelligence, not replacing it.

In conclusion, Generative AI has become a technological innovation in my life. It has helped me understand difficult concepts, explore new technologies, solve technical challenges, and grow as a student. When used responsibly, it is not simply a tool for obtaining answers, it is also a companion that encourages curiosity, continuous learning, and personal growth. As technology continues to evolve, I believe the most powerful combination will always be human creativity working together with artificial intelligence. Where artificial intelligence act as a companion, search engine and provides assistance.

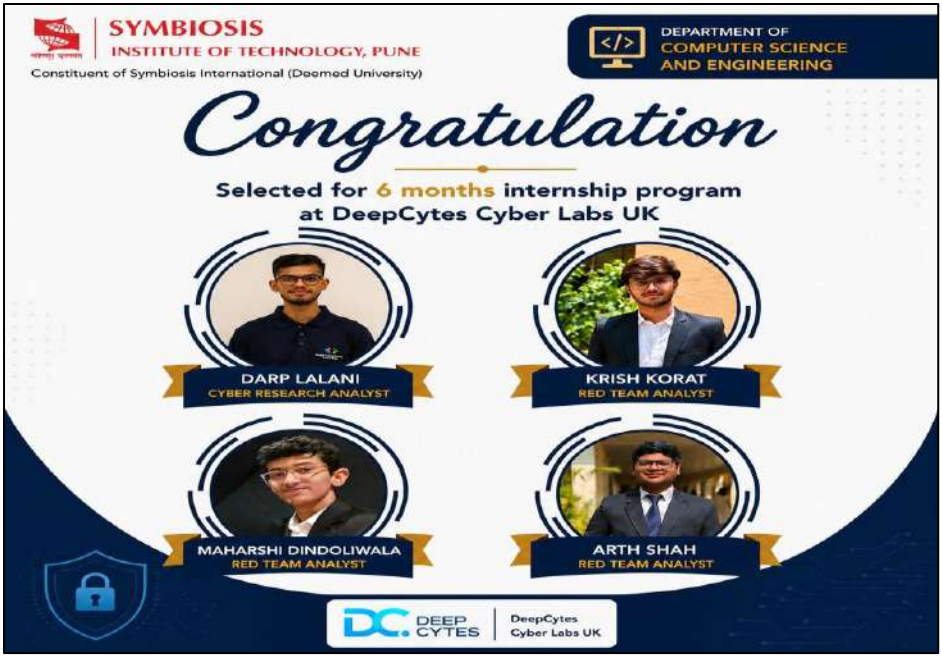
STUDENT ACHIEVEMENTS

Cybersecurity Fellowship Program:

Around 20 students selected for the 6-Month Fellowship Program 2026.

This initiative is brought to you by DARC Lab and CBC Club.





IEEE SA Cybersecurity Hackathon 2026:

Karan Desai, Pankhuri Varshney, and Reeti Agarwal secure 1st Place at the IEEE SA Cybersecurity Hackathon 2026.



Yash Parkhi secured the First Prize at the Defence Technology Expo 2026



Sai Ranadive, a Second-Year student from the department, has been awarded the Certificate of Merit (State Level) at the prestigious Design for Bharat – Youth Innovation Challenge 2026.



Excellent Performance in **CMR Hackathon 3.0**



Devaki Joshi, for completing the ACM Winter School on "Responsible AI" at XIM University, Bhubaneshwar



Armaan Jeswani awarded the **Bank of Baroda Achievers Award** in the Best All Rounder category.



Reeti Agarwal cleared the eWPT (Web Application Penetration Tester) certification by **INE Security**



Vinamra Jain cleared the Junior Penetration Tester certification by **INE Security**



DEPARTMENT HIGHLIGHTS

Patent Publication

1. **Dr. Sonali Kothari, Dr. Pooja Bagane, Prof. Amol Kamble**

   Topic: Secure Platform and Method for Non-human Identity Management with Encrypted Credential

2. **Dr, Sudipta Banerjee, Dr. Deepali Vora, Dr. Nilima Zade**

   Topic: Biometric and RFID-Based Intelligent Library Management System with Machine Learning Integration

3. **Dr, Ranjeet Bidwe, Dr. Sashikala Mishra**

   Topic: Autonomous Landmine Detection Robot Using GPR, Metal Detection, IoT and Remote Operation

   Topic: A WEB-BASED VIRTUAL COLLABORATION SYSTEM WITH PROXIMITY-BASED COMMUNICATION

4. **Dr. Sudipta Banerjee, Dr, Ranjeet Bidwe, Dr. Sashikala Mishra**

   Topic: Autonomous Landmine Detection Robot Using GPR, Metal Detection, IoT and Remote Operation

5. **Dr. Deppali Vora**

   Topic: Occupancy Driven Eenergy Management System Using MMWAVE RADAR and Cloud AI

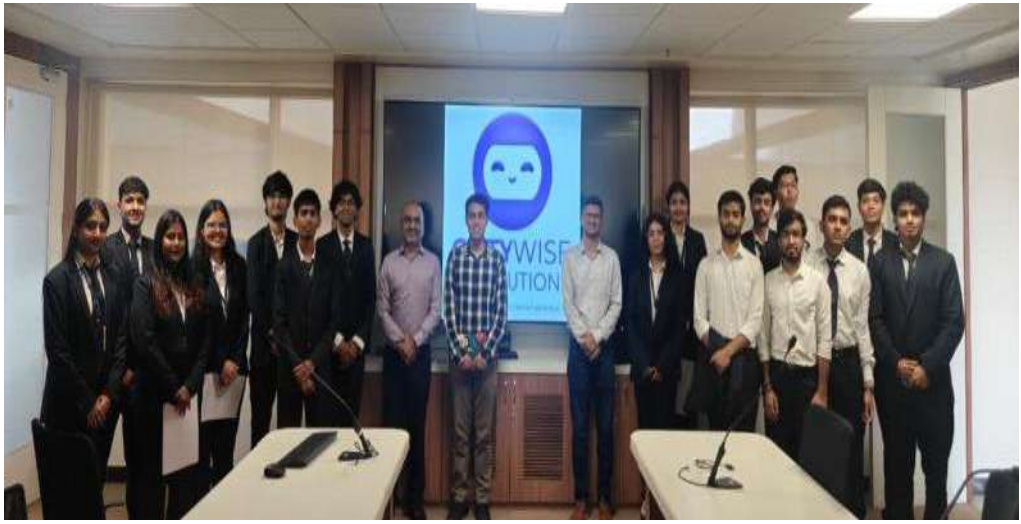
Expert session on "**Artificial Intelligence: The Mind of the Machine**" by Dr. Paulo Oliva from Queen Mary University of London on 2026



The Department was delighted to be a part of the recent **Industry–academia interaction with Kohler** Co. held on April 2026.



Department of Computer Science and Engineering (CSE) at Symbiosis Institute of Technology (SIT), Pune successfully hosted a campus **Placement Drive** for **Optywise.**



The inauguration of **the DARC Lab** in collaboration with DeepCytes Cyber Labs (UK)



Industry Conclave – NextGen Nexus 2026

The Department of Computer Science and Information technology Engineering, SIT Pune successfully organized the Industry Conclave on 11th April 2026.



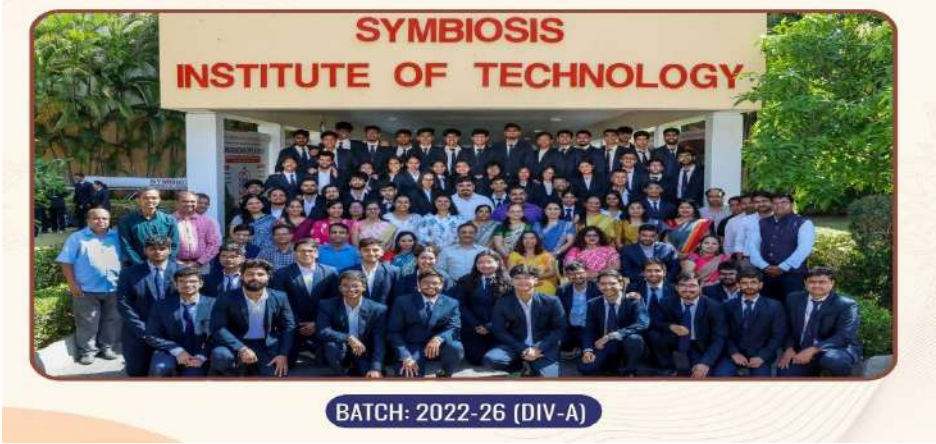


Department of Computer Science and Information technology Engineering, SIT Pune, successfully organized an IEEE Women's Day Panel Discussion on "Empowering Women in Engineering & Innovation" on March 2026.

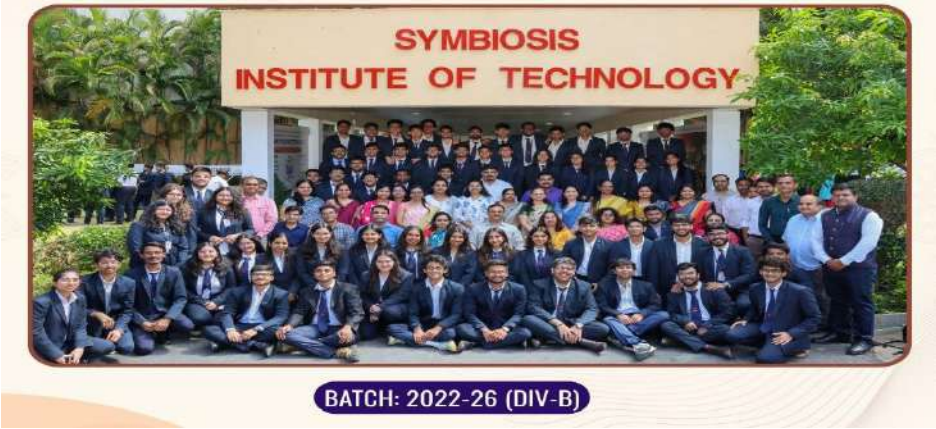




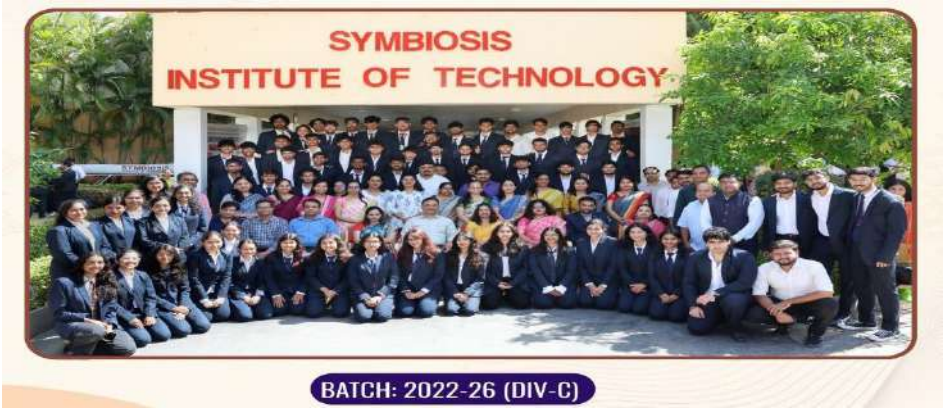
🎓 A Memorable Farewell for Batch 2022-26 🎓



BATCH: 2022-26 (DIV-A)

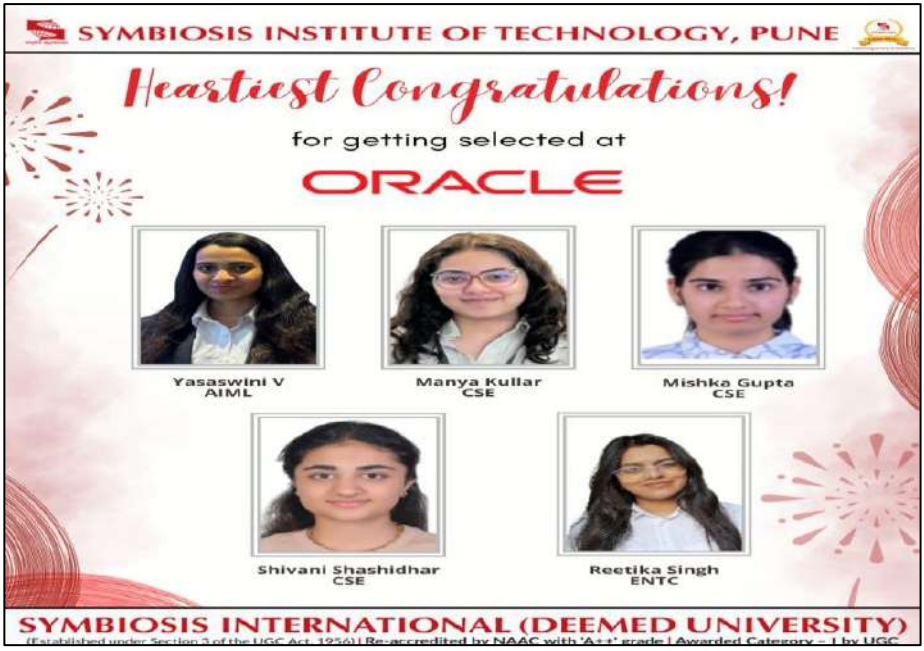


BATCH: 2022-26 (DIV-B)



BATCH: 2022-26 (DIV-C)

PLACEMENTS


SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE
Heartiest Congratulations!
for getting selected at
ORACLE
Yasaswini V
AIML
Manya Kullar
CSE
Mishka Gupta
CSE
Shivani Shashidhar
CSE
Reetika Singh
ENTC
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for getting selected at
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AIML
Kavya Kumar
ENTC
Dhairya A. Mehra
CSE
Yuvraj Aryan
CSE
Sanusha Menon
ENTC
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Heartiest Congratulations!

for getting selected at

Deloitte.



Armaan Jeswani
CSE



Krish Hemant Panchal
CSE



Aarya Yogesh Patil
CSE



Vithalani Gehna Vishal
CSE



Jay Vipulkumar Gondaliya
CSE

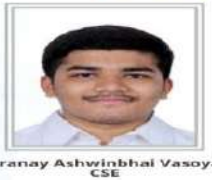
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Heartiest Congratulations!

for getting selected at

AMGEN



Pranay Ashwinbhai Vasoya
CSE



Sneha Sachidanand Padgaonkar
CSE



Shreyash Satish Govindwar
CSE

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SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
BATCH OF 2022-26

Congratulations students for getting selected at

MAERSK



Saumay Jaiwsal



Udayraj Jhala



Dhaval Bhimani

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
BATCH OF 2022-26

Congratulations students for getting selected at

pi-labs



Atharv Raje



Aryan Bachute

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SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

DEPARTMENT OF TRAINING & PLACEMENT

Congratulations

For being selected at Michelin India



Ankush Dutta
CSE



Ojas Verma
CSE



Devank Upadhyaya
CSE

BTech Batch 2023-27

SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

DEPARTMENT OF TRAINING & PLACEMENT

Congratulations

For being selected at Tracelink India



Drishti Mundhara
CSE



Shanvi Srivastava
CSE



Yug Choubey
CSE



Manasvi Pawa
CSE

BTech Batch 2023-27

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SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

DEPARTMENT OF TRAINING & PLACEMENT

Congratulations

For being selected at ZS Associates


Ananya Poundrik
CSE


Anum Agrawal
CSE


Arnav Karole
CSE


Arunabha M
CSE


Raghav Dhoot
CSE

BTech Batch 2023-27

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DEPARTMENT OF TRAINING & PLACEMENT

Congratulations

For being selected at ZS Associates


Sameer Shekhar
CSE


Aryan Sheladia
CSE


Saumya Soni
CSE


Yashraj Shrivastv
CSE


Riddhi Poddar
AIML

BTech Batch 2023-27

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Conclusion

As we come to the end of this edition of **Symbi-Sains**, we celebrate not just another academic year, but a year of learning, innovation, collaboration, and achievement in the Department of Computer Science and Engineering.

The pages of this magazine reflect the diverse journey of our students and faculty from exploring emerging technologies and sharing technical perspectives to participating in hackathons, innovation challenges, industry interactions, internships, projects, and research activities. These experiences demonstrate that education extends far beyond classrooms and laboratories.

Computer Science and Engineering is a rapidly evolving discipline, and staying relevant requires continuous learning, curiosity, adaptability, and the courage to experiment. The achievements and contributions featured in this edition remind us that meaningful progress comes from combining knowledge with creativity, technology with responsibility, and learning with action.

We are proud of the accomplishments of our students and faculty and grateful to everyone who contributed to the academic and co-curricular growth of the department. The magazine reflects our collective efforts and the vibrant culture of learning and innovation that defines our CSE community.

As we look ahead, let us continue to learn, explore, innovate, and create solutions that make a meaningful impact on society.

The journey does not end with this edition it inspires the next chapter**.**

Keep learning. Keep building. Keep innovating.